## Delivery-chain renderer specification

This appendix records the implementation details intentionally omitted from the scenario-level description in the main paper: the exact rendering budget, template inventory, parameter pools, and waveform-level operators. The released configuration and scripts are authoritative for reproducing the benchmark.

### Sampling policy

Each parent receives one direct control and a nominal budget of up to four children from each non-direct family before post-render validation. Table 8 lists the exact family inventory; failed validation removes a small number of candidate outputs from the final release. The deterministic per-parent template shuffle, sampling without replacement, and parameter-matched order pairs are described in the main paper.

Table 8: Nominal family-level budget before validation.

| Family | Templates | Children/parent |
|---|---|---|
| Direct | 1 | 1 |
| Platform-like | 6 | up to 4 |
| Telephony | 10 | up to 4 |
| Simulated replay | 7 | up to 4 |
| Hybrid | 9 | up to 4 |

### Template inventory and parameter pools

The released configuration contains 33 templates and 26 distinct ordered operator sequences; repeated multisets support controlled order comparisons. Table 9 gives the exact inventory.

Table 10 summarizes the fixed parameter pools and export contract.

### Operator realization

Every operator is executed as a waveform transform and its realized parameters are stored in the manifest trace.

**Resampling.** ffmpeg uses the SoX resampler at precision 28. Supported one-way conversions are 16k→8k/24k/32k and 8k/24k/32k→16k; round trips apply two successive conversions through a temporary waveform.

**Band-limiting.** The narrowband profile applies second-order high/low-pass filters at 250/3400 Hz followed by companding (0.01 s attack, 0.15 s decay). The wideband profile uses 50/7000 Hz cutoffs.

**Codec and re-encoding.** AAC, Opus, GSM, $\mu$-law PCM, and A-law PCM are realized as true ffmpeg encode–decode round trips. The codec operator follows its template directly. Re-encoding uses the most recent codec in *same* mode and a different codec from {AAC, Opus} in *cross* mode; absent prior context, simulated replay defaults to AAC and the relevant hybrid templates to Opus. GSM is encoded and decoded at 8 kHz.

**Packet loss.** Audio is divided into 20 ms frames. For target loss rate $p \leq 0.95$ and average burst length $b$, the two-state process uses

$$\begin{aligned} P(\mathrm{bad} \to \mathrm{good}) &= \min(1, 1/b), \\ P(\mathrm{good} \to \mathrm{bad}) &= \min\left(1, \frac{p\, P(\mathrm{bad} \to \mathrm{good})}{1-p}\right). \end{aligned} \tag{1}$$

Lost frames use repeat-fade, interpolation, or noise-fill concealment before concatenation.

**Room impulse responses.** Pyroomacoustics first derives absorption by inverse Sabine, samples valid source/microphone positions, generates and truncates the RIR, and convolves it with the waveform. A deterministic synthetic fallback uses a direct path, exponentially decaying random reflections, and a low-level noise floor. Outputs are peak-normalized after convolution.

**Additive noise.** White, pink, brown, hiss, hum, and synthetic babble are calibrated to the target SNR. Pink and brown use spectral shaping and integration, respectively; hum combines 50/100/150 Hz tones; babble sums six shifted and scaled streams. Babble, hum, and hiss use a non-stationary amplitude envelope.

**Call paths.** The composite pipeline applies profile-specific filtering/resampling, codec round trip, burst loss, frame-wise jitter-buffer simulation, and AGC/limiting. Each frame is displaced by a uniform offset in $[-J, J]$, overlap-averaged, and trimmed or padded to the original length.

## Release artifacts and protocol reconstruction

The package provides global and split-specific metadata CSVs, a dataset summary, dropped-row logs, and checks for speaker disjointness, duplicate identifiers/paths, format compliance, and parent coverage. The main paper lists the protocol-critical row fields. Protocol groups are derived deterministically: operator substitution fixes parent, label, family, and local context while changing one operator identity; parameter perturbation fixes identity and order while varying one parameter axis; order swap fixes the realized multiset and differs by one adjacent transposition; and lineage groups fix parent, label, and family, choose the shortest observed signature as reference, and compute graph distances over observed atomic edits. Thus the released metadata and grouping code reconstruct the benchmark without manually enumerated pair lists.

## Preservation-validation outputs

The main paper reports the validation backends, normalization rules, and family-wise DNSMOS, ASR, speaker, duration, and RMS results. The released evaluator runs from the packaged metadata (the test split by default) and writes row-level records keyed by `sample_id` and `parent_id`. Each record includes raw and normalized parent/child ASR hypotheses, WER/CER and their child-minus-parent drift, speaker cosine, duration ratio, RMS and peak-amplitude drift, the detection label, and explicit status/error fields. It also emits family-language aggregates and status counts, enabling percentile or label-conditioned analyses without rerendering. The paper pools bona fide and spoof rows because these statistics characterize delivery preservation rather than detection class.

Table 9: Template inventory as ordered operator blueprints before parameter instantiation.

| Family | Template ID | Ordered operator sequence |
|---|---|---|
| Direct | direct_clean | Identity chain with no operator. |
| Platform-like | aac_single | codec |
| | opus_single | codec |
| | aac_reencode | codec → reencode |
| | opus_reencode | codec → reencode |
| | aac_resample_reencode | codec → resample → reencode |
| | resample_opus | resample → codec |
| Telephony | nb_mulaw | bandlimit → codec |
| | nb_gsm | bandlimit → codec |
| | wb_opus | bandlimit → codec |
| | nb_mulaw_plr | bandlimit → codec → packet_loss |
| | nb_resample_mulaw_plr | resample → bandlimit → codec → packet_loss |
| | wb_resample_opus_plr | resample → bandlimit → codec → packet_loss |
| | wb_opus_resample_return | bandlimit → codec → resample |
| | session_nb_mulaw | call_path |
| | session_nb_gsm | call_path |
| | session_wb_opus | call_path |
| Simulated replay | rir_only | rir |
| | rir_noise | rir → noise |
| | noise_rir | noise → rir |
| | rir_reencode | rir → reencode |
| | reencode_rir | reencode → rir |
| | rir_noise_resample | rir → noise → resample |
| | resample_rir_reencode | resample → rir → reencode |
| Hybrid | opus_plr_rir | codec → packet_loss → rir |
| | rir_aac | rir → codec |
| | aac_rir | codec → rir |
| | bandlimit_codec_rir | bandlimit → codec → rir |
| | rir_bandlimit_codec | rir → bandlimit → codec |
| | rir_reencode_plr | rir → reencode → packet_loss |
| | reencode_rir_plr | reencode → rir → packet_loss |
| | resample_codec_rir | resample → codec → rir |
| | bandlimit_resample_codec | bandlimit → resample → codec |

## Benchmark-specific robustness metrics

We give exact definitions for the non-standard matched-pair and lineage metrics; standard EER, AUC, accuracy, and $F_1$ follow their usual definitions.

### Notation and thresholding

Let score $s_i \in \mathbb{R}$ increase with bona fide evidence and label $y_i \in \{0, 1\}$ use 1 for bona fide. For the pooled evaluation set $\mathcal{E}$, one task-level reference threshold is fixed as

$$\tau_{\text{ref}} = \begin{cases} \tau_{\text{EER}}(S_{\mathcal{E}}, Y_{\mathcal{E}}), & \text{if both classes occur,} \\ 0.5, & \text{otherwise,} \end{cases} \qquad (2)$$
$$\hat{y}_i = \mathbb{I}[s_i \geq \tau_{\text{ref}}].$$

The same threshold is reused across families, matched pairs, and lineages within a task; it is not recalibrated per subgroup. Accordingly, hard-decision metrics are common-operating-point diagnostics, whereas EER/AUC and MNSD are threshold-free.

### Matched-pair metrics

For operator substitution, parameter perturbation, and order swap, let $\mathcal{P} = \{(i_m, j_m)\}_{m=1}^{M}$ be the valid pairs whose members share a label and differ only in the targeted local intervention.

**Pair consistency rate.**

$$\text{PCR} = \frac{1}{M} \sum_{m=1}^{M} \mathbb{I}[\hat{y}_{i_m} = \hat{y}_{j_m}], \qquad (3)$$

PCR measures decision stability under the controlled edit, regardless of correctness.

**Pair joint accuracy.**

$$\text{PJA} = \frac{1}{M} \sum_{m=1}^{M} \mathbb{I}[\hat{y}_{i_m} = y_{i_m} \ \wedge \ \hat{y}_{j_m} = y_{j_m}], \qquad (4)$$

PJA requires both members to be correct.

Table 10: Fixed parameter pools and export constraints.

| **Group** | **Released range or set** |
|---|---|
| Codec bitrates | AAC {24, 32, 48}; Opus {16, 24, 32}; telephony Opus {16, 24}; re-encode {24, 32} kbps |
| Resampling | Family-specific one-way and round-trip 8/16/24/32 kHz conversions |
| Re-encoding | AAC or Opus with same/cross selection |
| Packet loss | Loss {1, 3, 5, 10}%; burst {2, 3, 5} frames; three concealment modes |
| Call path | Jitter {0, 8, 16} ms; AGC {mild, telephony} |
| Noise | Six synthetic types; SNR {30, 20, 15, 10} dB |
| RIR | RT60 {0.2, 0.4, 0.6, 0.8} s; distance {0.5, 1.0, 2.0, 3.0} m; three rooms |
| Export | Mono 16 kHz PCM WAV; duration in [1, 30] s |

**Mean normalized score drift.** Let $D_{\mathcal{E}}$ be the pooled-score normalization

$$D_{\mathcal{E}} = \begin{cases} \mathrm{IQR}(S_{\mathcal{E}}), & \mathrm{IQR}(S_{\mathcal{E}}) > 10^{-12}, \\ 1, & \text{otherwise.} \end{cases} \tag{5}$$

Then

$$\mathrm{MNSD} = \frac{1}{MD_{\mathcal{E}}} \sum_{m=1}^{M} |s_{i_m} - s_{j_m}|, \tag{6}$$

which measures score sensitivity independently of the hard decision.

**Symmetric misclassification risk (auxiliary).** Define sample-level 0–1 loss as

$$\ell_i = \mathbb{I}[\hat{y}_i \neq y_i]. \tag{7}$$

The corresponding pairwise quantity is

$$\mathrm{SMR} = \frac{1}{M} \sum_{m=1}^{M} \frac{\ell_{i_m} + \ell_{j_m}}{2}. \tag{8}$$

SMR is the average pairwise misclassification burden.

## Delivery-lineage metrics

A lineage groups samples with the same parent, family, and label. Each node $v$ carries an observed operator signature $\sigma(v)$; the shortest observed signature is the reference $r_g$.

**Lineage graph and node correctness.** Nodes are observed signatures; an undirected edge joins signatures that differ by one parameter perturbation, operator substitution, adjacent swap, or insertion/deletion. Let $d_g(v)$ be the shortest-path distance from $r_g$. Node correctness is

$$c(v) = \mathbb{I}[\hat{y}(v) = y(v)]. \tag{9}$$

**Censored first-failure distance.** For lineage $g$,

$$\rho_g = \min\{d_g(v) : c(v) = 0\}, \tag{10}$$

when a failure is observed; otherwise the lineage is right-censored.

Let $D_{\max} = \max_{g \in \mathcal{G}} \ \max_{v \in g} d_g(v)$ be the largest observed lineage distance in the evaluation set. The censored first-failure distance is then

$$\tilde{\rho}_g = \begin{cases} \rho_g, & \exists v \in g : \ c(v) = 0, \\ D_{\max} + 1, & \text{otherwise,} \end{cases} \tag{11}$$

and the reported aggregate is

$$\text{C-FFD} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \tilde{\rho}_g. \tag{12}$$

Higher C-FFD means correctness persists farther from the reference.

**Robust-at-distance.**

$$R_k = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \mathbb{I}[\forall v \in g : \ d_g(v) \leq k \Rightarrow c(v) = 1], \quad k = 0, 1, \ldots, D_{\max}. \tag{13}$$

$R_k$ is the fraction of lineages entirely correct through distance $k$.

**AURC-chain.**

$$\text{AURC-chain} = \frac{1}{D_{\max} + 1} \sum_{k=0}^{D_{\max}} R_k. \tag{14}$$

AURC-chain averages the complete robustness profile; higher is better.